Human Capital Theory

EC306- Wages and Employment

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Goals of This Section

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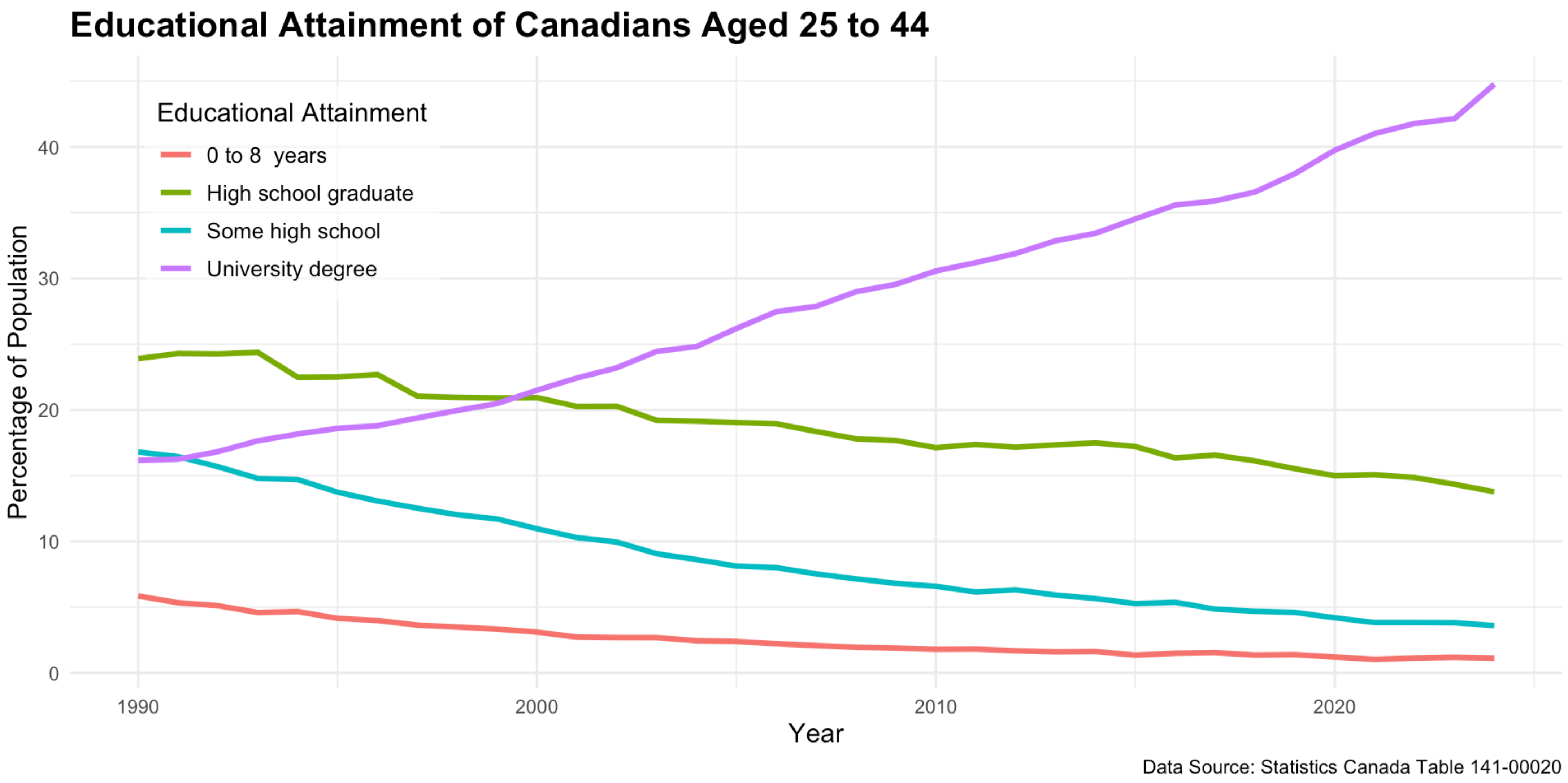
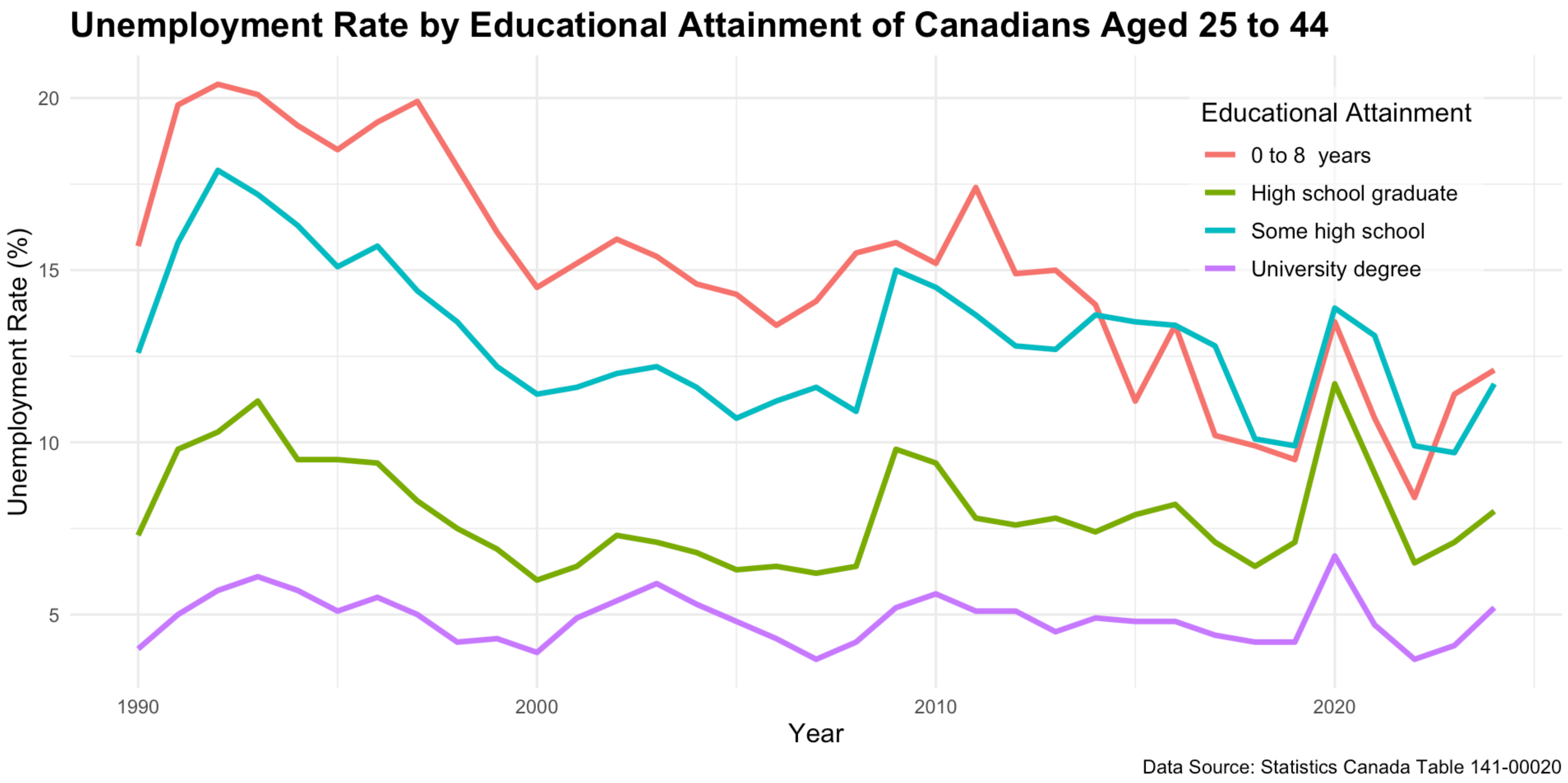
- Bring together demand and supply to determine wages and employment
- Discuss the effect of market power in goods market on equilibrium wages and employment
- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Analyze minimum wages

Introduction

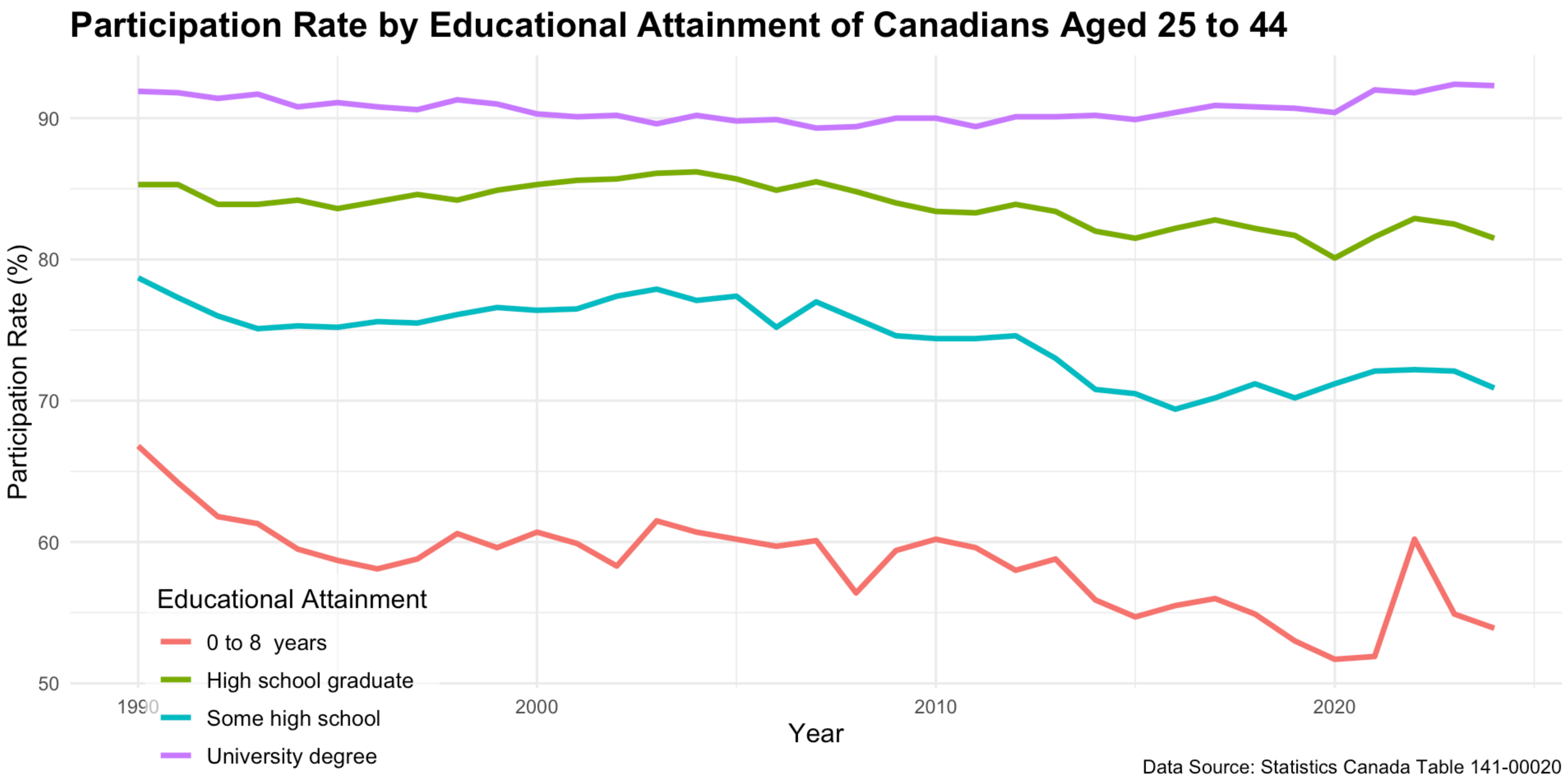
Educational Attainment



Unemployment Rate



Participation Rate



Human Capital Theory

Human Capital Theory

- Firms often invest in physical capital (e.g., machines, tools, buildings)
  - Incur some cost today to invest in capital
  - Boosts the productive ability of workers
  - Increases the output flow in the future
- Decision to invest is made by comparing costs to benefits
  - Invest if benefits outweigh costs
  - Costs are usually upfront, while benefits occur slowly over time

Human Capital Theory

- Individuals make similar investments in *human* capital
  - **Human Capital:** An individual's stock of productive skills, knowledge, and abilities
  - People incur upfront costs for education, training, etc.
  - Training improves productivity, and therefore income
  - Individuals earn higher income streams in the future
- We will study investments in human capital in a similar way to physical capital

Human Capital Theory

- Human capital investments involve some special considerations
  - Non-wage benefits
    - Consumption value
    - Differences in risk of unemployment in downturns
  - It is more difficult to finance education investments
    - No collateral
- Social versus private costs/benefits
  - Private costs and benefits are specific to the individual
  - But there may be social costs and benefits as well
    - E.g., positive externalities from education

Private Investments In Education

Introduction

- The human capital investment decision boils down to comparing lifetime income streams
  - Different streams of income are associated with different levels of schooling
  - Higher levels of schooling usually lead to higher income streams
- We will outline a process to decide which stream to choose

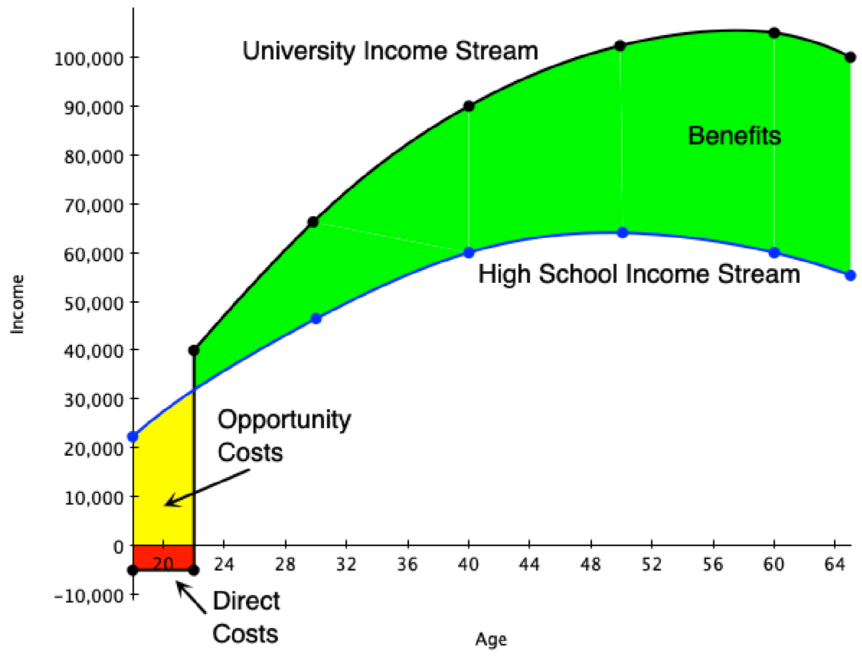
To make the model tractable, we assume:

1. No direct utility or disutility from schooling
2. Hours of work are held fixed
3. Certainty of income streams
4. Individuals can borrow and lend at interest rate r

Attending University

- Suppose we want to analyze whether it is financially worthwhile to attend university
  - Start from the perspective of an 18-year-old high school graduate
  - This student is deciding whether or not to attend university
  - Students who do not attend university move immediately into the workforce
- The graph on next slide depicts two income streams:
  - **Black line:** university path
  - **Blue line:** high school path
  - **Red and yellow:** costs of attending university
  - **Green:** benefits of university education

Attending University



- Direct costs are in red
  - Books, fees, tuition
- Foregone wages are in yellow
  - Income not earned during the time in university
- Net gains in wages are in green
  - Earnings of university grad above a high school grad
- Beneficial to attend if present value of green exceeds present value of yellow plus red

Attending University

- People will choose the level of schooling that maximizes the net present value of lifetime earnings
- We illustrated that graphically above
- Mathematically, express the present value of earnings for a high school graduate as

$$PV = \frac{Y}{(1+r)^0} + \frac{Y}{(1+r)^1} + \ldots + \frac{Y}{(1+r)^{T-18}}$$

- In this equation
  - Y = annual earnings of high school graduate
  - r = interest rate
  - T = expected retirement age

Attending University

- We can approximate the present value of earnings for a high school graduate as

$$PV \approx Y + \frac{Y}{r}$$

- Approximation works if T is long enough
- Now consider a person who attends one year of university
- They incur direct costs D in the first year and earn no income in that year
- Every year they work, they gain an amount ΔY above the high school graduate

Attending University

- Their income stream is

$$PV^* = \frac{-D}{(1+r)^0} + \frac{Y + \Delta Y}{(1+r)^1} + \ldots + \frac{Y + \Delta Y}{(1+r)^{T-18}}$$

- This can be approximated as

$$PV^* \approx -D + \frac{Y + \Delta Y}{r}$$

- The net present value of attending this additional year of university is

$$NPV = PV^* - PV \approx -(D + Y) + \frac{\Delta Y}{r}$$

Attending University

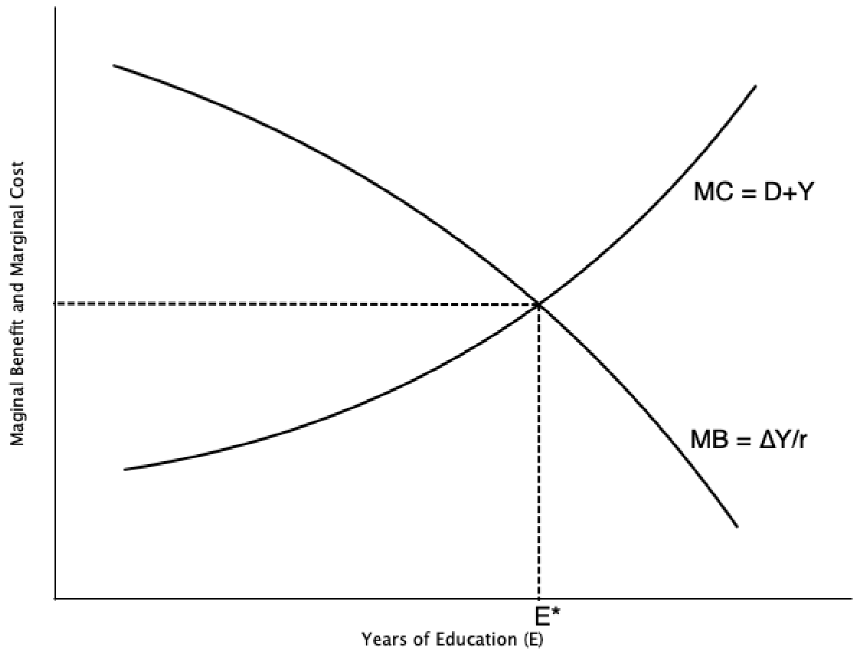
- The term $D + Y$ is the marginal cost of attending university for one year
- The term $\frac{\Delta Y}{r}$ is the marginal benefit of attending university for one year
- Attending university for one year is worthwhile if

$$NPV > 0 \implies \frac{\Delta Y}{r} > D + Y$$

- A person will continue to attend more years of schooling until $MB = MC$, or

$$\frac{\Delta Y}{r} = D + Y$$

Attending University



- Graph on left illustrates optimal schooling decision
- Marginal benefit curve is downward sloping
  - Each additional year of schooling yields a smaller increase in wages
- Marginal cost curve is upward sloping
  - Each additional year of schooling incurs higher direct costs, foregone wages, psychic costs
- Optimum is where MB = MC

Attending University

- Another way to look at the schooling decision is through the rate of return to schooling
- This method uses the concept of the internal rate of return (IRR)
- The IRR is the interest rate that makes the NPV of an investment equal to zero
- In our setup, the IRR is the interest rate that satisfies

$$0 = -(D + Y) + \frac{\Delta Y}{IRR}$$

- Rearranging gives

$$IRR = \frac{\Delta Y}{D + Y}$$

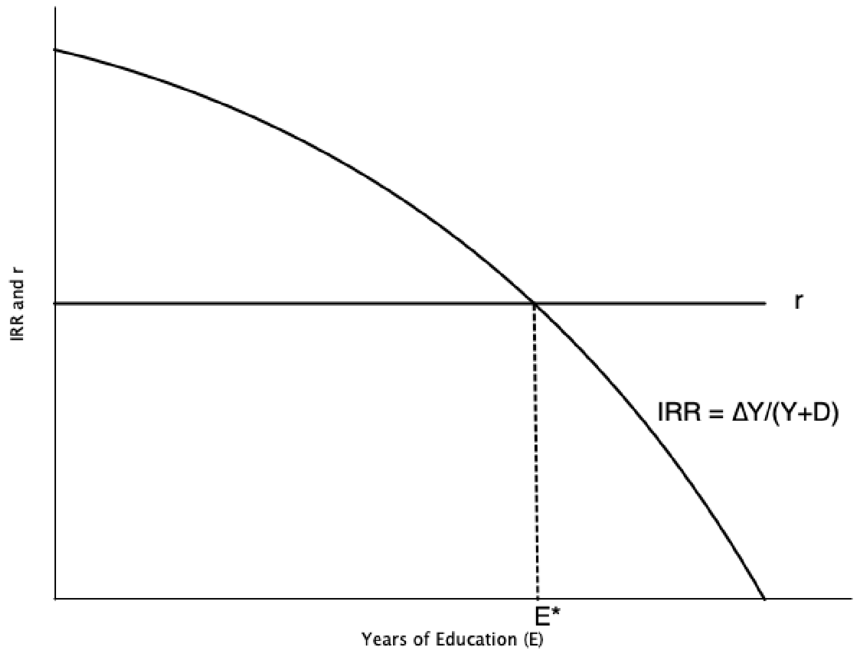
- The IRR is the percentage return on the investment in schooling

Attending University

- The alternative to attending university is to invest the money $D + Y$ at the market interest rate r
  - Opportunity cost of investing in schooling is r
- If the IRR exceeds r, then investing in schooling is worthwhile
- A person would continue to attend more years of schooling until

$$IRR = r \implies \frac{\Delta Y}{D + Y} = r$$

Attending University



- Graph on left illustrates alternative optimal schooling decision
- Market interest rate is r
- Internal rate of return IRR is decreasing in schooling
  - Each additional year of schooling yields a smaller increase in wages
- Optimum is where IRR = r
- Yields same optimal schooling level as previous method

Comparative Statics

- We can use this framework to see how changes in parameters and policies affect schooling decisions
- Suppose the direct costs D of attending university decrease
  - E.g., government introduces tuition subsidies
  - This decreases the marginal cost of schooling
  - Leads to an increase in optimal schooling level
- Suppose the market interest rate r decreases
  - E.g., due to monetary policy
  - This increases the present value of future earnings
  - Leads to an increase in optimal schooling level

Comparative Statics

- What if the income tax system becomes more progressive?
  - Higher taxes on high income earners
  - Decreases the after-tax increase in earnings ΔY
  - Leads to a decrease in optimal schooling level
- Suppose the pay for jobs in the oil sector increases substantially
  - E.g., due to a boom in oil prices
  - Increases the earnings of high school graduates Y
  - Leads to a decrease in optimal schooling level

Complications

- Information asymmetries
  - Students may not know the true costs and benefits of schooling
  - May lead to under- or over-investment in schooling
- Disutility of schooling
  - Students may dislike attending school
  - This adds to the marginal cost of schooling
- Different degrees have different returns
  - Some fields of study lead to higher earnings than others
  - Students must choose not only how much schooling, but what type

Complications

- A major complication is financing/credit constraints
  - Many students cannot afford the upfront costs of schooling
  - May not be able to borrow against future earnings
  - Leads to under-investment in schooling compared to optimal level

Signalling

Introduction

- In the Human Capital model, there are key assumptions that make the model work:
  - Education increases productivity
  - Productivity is perfectly observed by employers
  - Wages are paid according to productivity
- The **signalling model** emphasizes an alternative view:
  - Education may not increase productivity
  - Instead, it reveals productivity to employers
  - It acts as a signal of worker ability
  - People only invest in education to signal their ability to employers

Model

- The basics of the model are
  - Employers do not observe productivity until after a worker is hired
  - Employees know their own ability but find it difficult to credibly reveal it to employers
  - Without knowing productivity, firms pay employees according to *average* productivity in the population
  - Under certain conditions, obtaining education can serve as a *signal* of productivity, allowing workers to be paid accordingly

Model

- For education to actually be informative about ability:
  - It must be costly to acquire
  - It must be **sufficiently more costly** for low-ability individuals than for high-ability individuals
    - Otherwise, low-ability workers might find it profitable to *fake* high ability by acquiring education
  - Employers' beliefs must be confirmed in the future
    - Otherwise, they will update their wage offers
- Given the conditions above, education helps solve an information problem:
  - It correctly reveals ability through educational attainment
  - High- and low-ability workers are then correctly paid according to their productivity

Model

- Below we do a more formal treatment of the model
- Assume that:
  - There are two types of workers: low ability (L) and high ability (H)
  - A fraction q are Type L, and $(1 - q)$ are Type H
  - For Type L workers:
    - Cost of education: $c_L(s) = s$
    - Marginal Product (MP): 1
  - For Type H workers:
    - Cost of education: $c_H(s) = \frac{s}{2}$
    - Marginal Product (MP): 2

Model

- Employees know their own ability, but employers do not
- Employers instead form *beliefs* about employee ability based on observed education levels
- What would happen if there were no way to signal ability?
- Firms would have no way of knowing who is high or low ability
  - So they offer everyone the same wage.
- The wage equals the **average productivity**:

$$\bar{w} = 1 \cdot q + 2 \cdot (1 - q) = 2 - q$$

Model

- Example: 50% Low-Ability Workers
  - If 50% of the population are low ability ($q = 0.5$):

$$\bar{w} = 2 - q = 1.5$$

- Recall that
  - Type L workers produce $MP = 1$
  - Type H workers produce $MP = 2$
- Therefore:
  - Type L is paid **more** than they produce
  - Type H is paid **less** than they produce
  - On average, total pay equals total production

Model

- Suppose employers observe education
- They form beliefs about how it relates to ability, and pay according to those beliefs.
- One possible set of employer beliefs:
  - Workers with education $s > s^*$ are Type H
  - Workers with education $s \leq s^*$ are Type L
  - $\boxed{\text{s^}}$ is some cutoff level of education
- Firms pay wages according to the believed type:

$$w = \begin{cases} 2, & \text{if } s > s^* \\ 1, & \text{if } s \leq s^* \end{cases}$$

Model

- In this model, workers decide how much education to acquire.
- A worker of type (t) maximizes utility:

$$U_t(s) = w(s) - c_t(s)$$

- All workers can obtain $U_t(s) = 1$ by acquiring **no education**.
    - Therefore, the payoff to acquiring any education must satisfy $U_t(s) \geq 1$
- Workers will never find it useful to obtain **more than** s^* years of education.
    - Doing so increases costs without increasing payoffs.
- Thus, the decision is binary: either **0** or s^* years of education.

Model

- Type H Workers

$$U_H(0) = w(0) - c_H(0) = 1 - 0 = 1$$

$$U_H(s^*) = w(s^*) - c_H(s^*) = 2 - \frac{s^*}{2}$$

- A Type H worker will acquire s^* years of education if:

$$U_H(s^*) > U_H(0)$$

$$2 - \frac{s^*}{2} > 1$$

$$s^* < 2$$

Model

- For Type L Workers

$$U_L(0) = w(0) - c_L(0) = 1 - 0 = 1$$

$$U_L(s^*) = w(s^*) - c_L(s^*) = 2 - s^*$$

- A Type L worker will acquire s^* years of education if:

$$U_L(s^*) > U_L(0)$$

$$2 - s^* > 1$$

$$s^* < 1$$

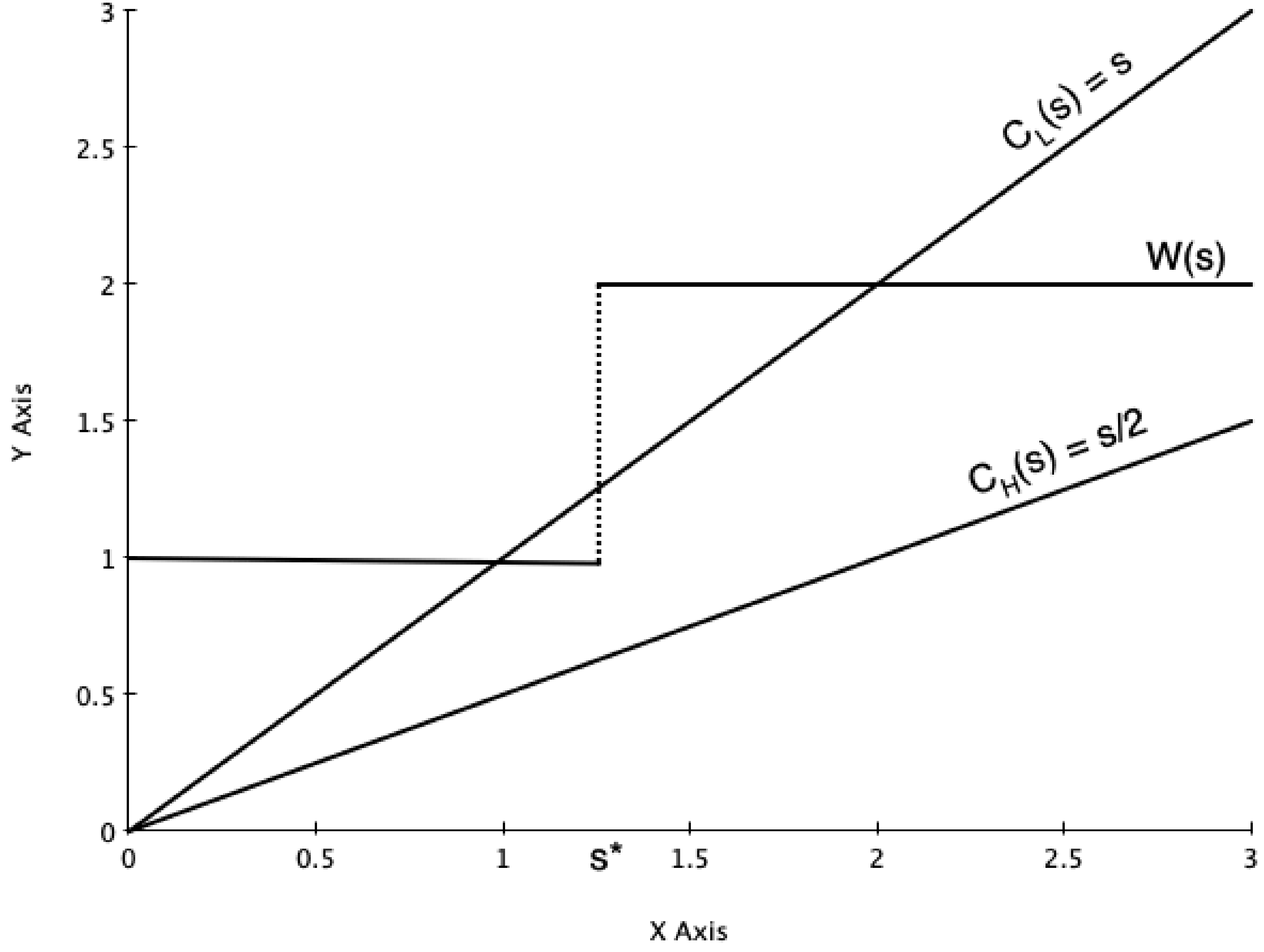
Model

- If $s^* < 1$:
  - Both Type L and Type H acquire education s^*
  - The signal is **not informative**
- If $s^* > 2$:
  - Both Type L and Type H acquire **no education**
  - The signal is **not informative**
- If $1 < s^* < 2$:
  - Type H obtains education s^*
  - Type L obtains no education
  - Education **reveals worker type**

Model

- Equilibrium
- Occurs when s^* lies **between 1 and 2**.
  - This is an equilibrium because the beliefs based on this strategy are **correct**
  - A firm can only maintain beliefs if they are **validated** by observed behaviour
- Any s^* between **1 and 2** will satisfy equilibrium conditions
  - The equilibrium with the **highest welfare** is the one closer to $s^* = 1$
- This is called a **separating equilibrium**.
  - When everyone gets the same level of education, it is a **pooling equilibrium**.
  - The case where there is **no education signal** is one example of a pooling equilibrium.

Model



Model

- Sometimes everyone is worse off under the separating equilibrium:
- Recall inn the pooling equilibrium, nobody gets education.
  - Everyone is paid $\bar{w} = 2 - q$
- Under the separating equilibrium:
  - Type L are paid \$1 — they are worse off unless $q = 1$
  - Type H are paid $2 - \frac{s^*}{2}$ — they are worse off unless $2 - q > 2 - \frac{s^*}{2} \rightarrow s^* > 2q$
- In our previous example where $q = 0.5$
  - Type H individuals are worse off for any $s^* > 1$

Model

- Key Ideas of the Signalling Model
- Depending on firm beliefs about the relationship between education and ability, it may be worthwhile to acquire education.
- There must be a difference in the cost of acquiring education for high- and low-ability workers.
- If firms' beliefs are not confirmed, they are not sustainable in equilibrium.

Estimating Returns to Schooling

Introduction

- We studied two theories of private education investment
  - These theories explain why individuals might invest in education
- Empirical economists estimate the size of the return using data
- We will study how empirical economists approach estimation
  - Mostly based on the **Human Capital Earnings Function**
- We then discuss results from existing research
  - Rates of return to additional education are large
  - Some evidence they are rising over time
  - Returns are mostly attributed to human capital, but there is some evidence of signalling effects

The Human Capital Earnings Function

- Most estimates of the return to education are based on variations of the Human Capital Earnings Function (also known as the Mincer Equation)

$$\ln Y = \beta_0 + \beta_1\, schl + \beta_2\, exp + \beta_3\, exp^2 + \epsilon$$

- The variables are
  - Y is some measure of earnings, income, or wages
  - $schl$ is years of schooling
  - exp is potential experience
    - $exp = age - schl - 5$
    - Used because actual experience is not always observed

The Human Capital Earnings Function

- The Mincer equation uses the natural logarithm of Y as the dependent variable
- When we use the natural log of the dependent variable, coefficients measure the percent change in the dependent variable per unit change in the independent variable

$$\beta_1 = \frac{\partial \ln Y}{\partial schl}$$

$$= \frac{\partial \ln Y}{\partial Y} \cdot \frac{\partial Y}{\partial schl}$$

$$= \frac{1}{Y} \cdot \frac{\partial Y}{\partial schl}$$

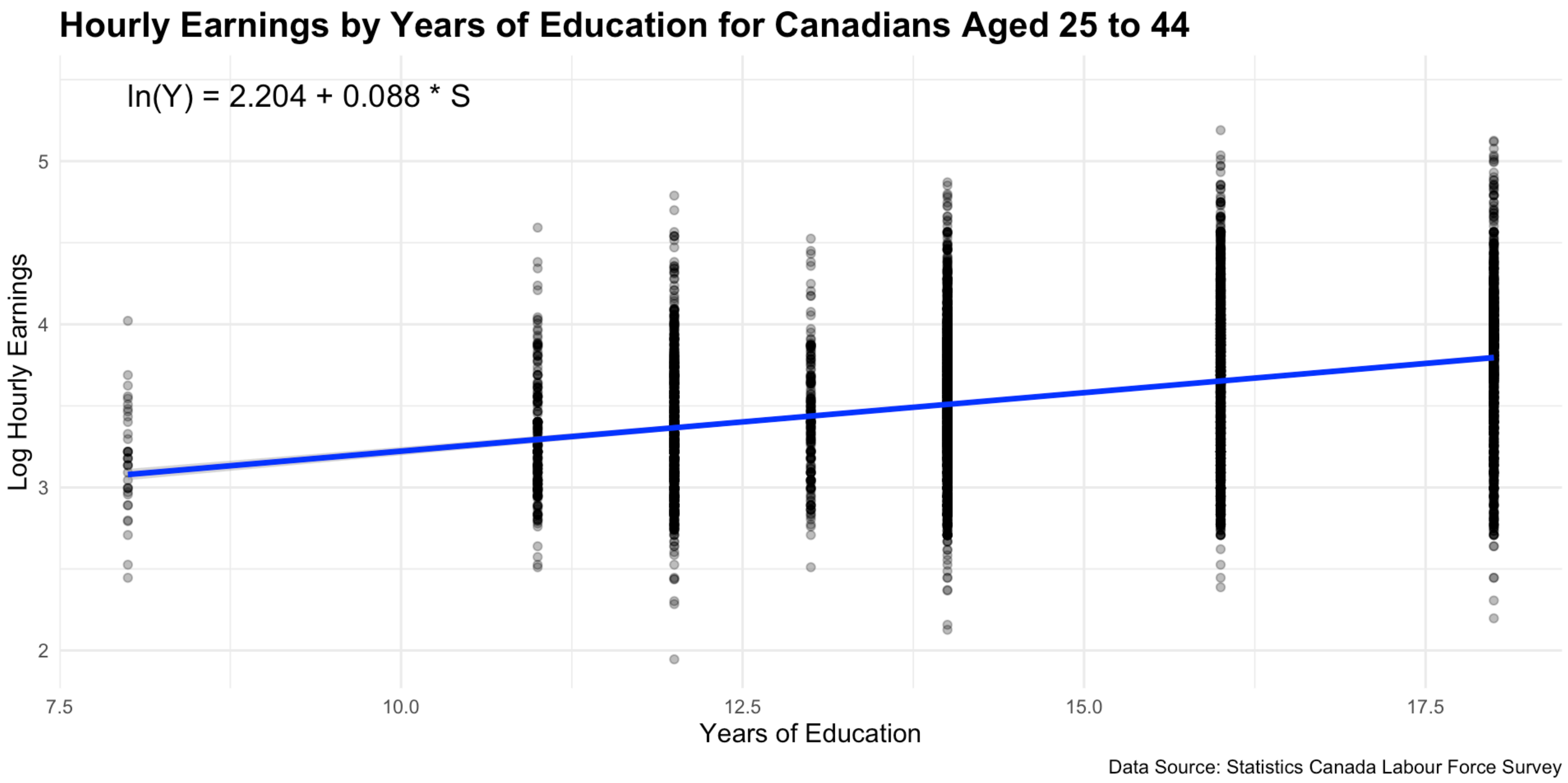
- This is the percent change in Y per additional year of schooling
- β_1 is interpreted as the IRR

The percent change in yearly earnings per additional year of schooling

The Human Capital Earnings Function

- One way to estimate this equation is to use Ordinary Least Squares (OLS) regression
- There are two main data sources in Canada for this
  - The Canadian Census
    - Large sample sizes
    - Detailed information on education and income
    - Only conducted every 5 years
  - The Labour Force Survey (LFS)
    - Monthly survey
    - Smaller sample sizes
    - Less detailed information on education and income

The Human Capital Earnings Function



The Human Capital Earnings Function

- Graph above shows data on schooling and hourly earnings from the LFS in September 2025 (black dots)
- Also shows the fitted OLS regression line (blue line)
- There is a clear positive relationship between schooling and earnings
- Issues with this graph
  - Does not control for experience
  - Education data not actually stored in years of schooling
    - Actually stored in categories (e.g., high school graduate, some university, university degree)
    - I converted categories to approximate years

The Human Capital Earnings Function

- The more accurate way is to
  - Run the full Mincer equation regression
    - Control for experience or age
  - Use categorical education variables instead of years of schooling
    - Need to omit one group
    - This forms the “reference category”
    - Coefficients on the education categories measure the percent difference in earnings relative to the reference category
- Slide below shows results from a Mincer equation regression using September LFS data

The Human Capital Earnings Function

	Men	Women
0–8 years	-0.125 (0.019)	-0.178 (0.024)
Some high school	-0.073 (0.010)	-0.103 (0.012)
Some postsecondary	0.065 (0.012)	0.076 (0.012)
Postsecondary certificate	0.163 (0.007)	0.141 (0.007)
Bachelor’s degree	0.296 (0.008)	0.366 (0.007)
Graduate degree	0.397 (0.009)	0.485 (0.009)
Num.Obs.	28560	28320
R2	0.268	0.296
RMSE	0.40	0.37

Ability Bias

- Major problem with estimating returns to schooling is **ability bias**
  - Higher ability people choose more schooling but would earn higher wages even without schooling
  - So OLS estimate reflects both the return to schooling and the return to ability
- Ability bias is thought to be positive
  - Higher ability people choose more schooling on average
  - This leads to overestimating the return to schooling
- We can illustrate this with a model

Ability Bias

- Suppose the true earnings equation is

$$\ln Y = \beta_0 + \beta_1 schl + \beta_2 abil + \epsilon$$

- s: schooling
- a: ability
- ϵ: other factors with $E[\epsilon \mid s, a] = 0$
- Assume schooling increases with ability through a linear relationship

$$abil = \alpha_0 + \alpha_1 schl + \upsilon$$

with υ uncorrelated with a

Ability Bias

- If we estimate a regression omitting ability

$$\ln Y = \tilde{\beta_1} schl + e$$

- The OLS coefficient on schooling equals

$$\tilde{\beta_s} = \frac{Cov(lnY, s)}{Var(s)}$$

Ability Bias

- Plug $\ln Y = \beta_1 schl + \beta_2 abil + \epsilon$ into formula for $\tilde{\beta_s}$

$$\tilde{\beta_s} = \frac{Cov(lnY, schl)}{Var(schl)} = \frac{Cov(\beta_0 + \beta_1 schl + \beta_2 abil + \epsilon, schl)}{Var(schl)}$$

- Because β_0 is constant and ϵ is uncorrelated with $schl$

$$\tilde{\beta_s} = \frac{\beta_1 Var(schl) + \beta_2 Cov(abil, schl)}{Var(schl)} = \beta_1 + \beta_2 \alpha_1$$

Ability Bias

- The term $\beta_2\alpha_1$ is the ability bias
- It depends on
  - β_2: the earnings return to ability
  - α_1: the relationship between ability and schooling
- If both terms are positive, then ability bias is positive
- This means that OLS overestimates the return to schooling
  - Estimates partly reflect the return to ability, partly the return to schooling

Addressing Ability Bias

- Economists have developed methods to address ability bias
  - **Twin Studies:** Compare earnings of identical twins with different schooling levels
    - Presumably, twins have same level of ability
    - Comparing them avoids ability bias
  - **Instrumental Variables (IV):** Method that isolates variation in schooling unrelated to ability
    - E.g., changes in compulsory schooling laws, proximity to university
    - These changes affect schooling but are unrelated to ability

Twin Studies

- Twin studies compare earnings of identical twins with different schooling levels
- Ashenfelter and Rouse (1998) use data on identical twins in the US
  - Attended Twins festival and collected data on schooling and earnings
  - Verified it with outside data
- Using OLS, they estimate a biased return to schooling of about 11% per year
- Using twin differences, they estimate a return of about 7% per year
  - Suggests ability bias of about 4 percentage points

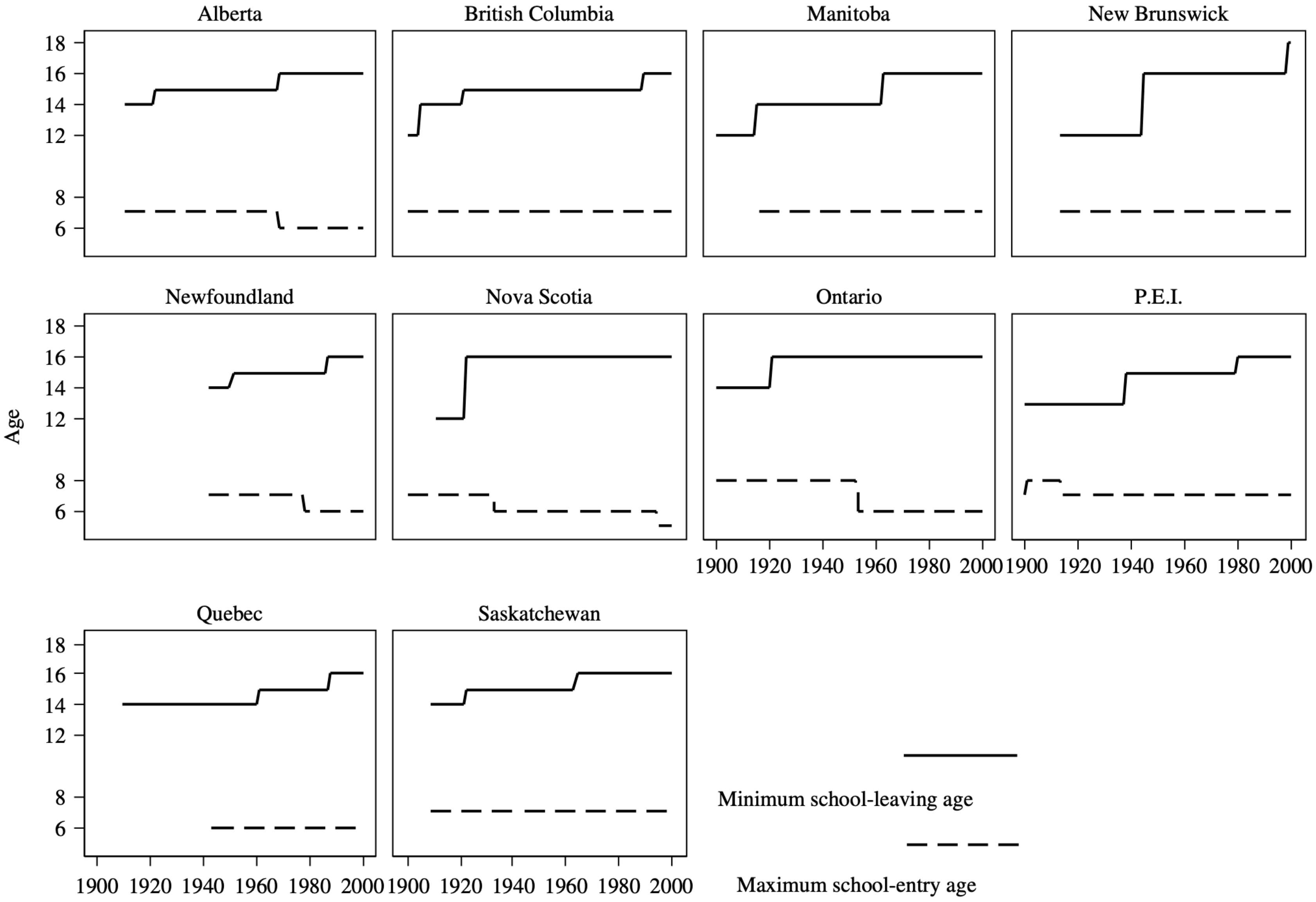
Compulsory Schooling

- Another method is to use changes in compulsory schooling laws as an instrument for schooling
- Several US states have a minimum school leaving age
  - Generally around 16 years old
- Means that students born earlier in the year can leave school earlier
  - Someone born in January can leave earlier than someone born in December of the same year
  - Means that on average people born later in the year have more schooling
- Angrist and Krueger (1991) use this variation to estimate returns to schooling
  - Estimate a return of about 7-10% per year of schooling

Compulsory Schooling

- Can use compulsory schooling laws in a different way
- Canada has maximum school entry laws and minimum school leaving laws
  - Over time, students mandated to enter school earlier and leave later
  - This increases average schooling levels, unrelated to ability
- Oreopoulos (2006) uses this variation to estimate returns to schooling in Canada
  - Finds that an additional year of compulsory schooling increases earnings by about 7-12%

Compulsory Schooling



Signalling

- All studies cited so far have attributed the returns to schooling to human capital accumulation
- Some studies have attempted to separate human capital from signalling effects
- Ferrer and Riddell (2002) use Canadian data to estimate credential effects
  - Look at people with same years of schooling, but different credentials
  - E.g., Someone with a high school diploma versus someone with same years but did not graduate
- Find that obtaining the credential does matter

Social Returns to Education

Social Returns to Education

- Most of the discussion so far has focused on private returns to education
  - Returns to individuals of obtaining more schooling
- We know there are spillovers to society as a whole
  - Positive externalities from education
- Studies show that
  - States with higher average education levels have higher earnings in excess of the private return
  - The number of highly educated immiggrants in STEM has a positive effect on productivity of other workers
  - More highly educated populations have lower crime rates, better health outcomes, more civic participation

Education and Inequality

Education and Inequality

- Earnings inequality has increased in many countries over the past few decades
- In Canada
  - Rose over the 80s, 90s
  - Has been stable since
- A couple of different sources for this
  - Returns to schooling
  - Growth in inequality among people with the same level of schooling
- In Canada, much of it is related to inequality within education groups
  - But still partly related to returns to schooling

Education and Inequality

Education and Inequality

Education and Inequality

- Why has the return to education increased?
  - **Skill Biased Technological Change**: New technologies increase demand for skilled workers
    - Increases wage premium for higher education
  - **Globalization**: Increased trade and outsourcing
    - Reduces demand for lower-skilled workers locally
  - **Institutional Changes**: Decline in unionization, changes in labor market policies
    - Negatively affects bargaining power of lower-skilled workers
- Skill biased technical change is thought to be major driver in the 80s

Training

Introduction

- Formal schooling is one way to develop human capital
- Another important way is through job training
  - ** General Training**: Skills that are useful at many firms
  - ** Specific Training**: Skills that are only useful at a specific firm
- Training helps develop skills that increase human capital
- In this section we evaluate the economics of job training

Who Pays for Job Training

- It is not uncommon for firms to pay for job training
  - Some companies provide training to all employees
  - Other will fund things like MBA
- Under what circumstance would the firm pay versus the individual?
- Differs for general versus specific training

Who Pays for Job Training

- General training increases productivity at many firms
  - Employees can take skills to other firms
- Firms will offer higher wages to employees with general training
  - Very similar to schooling
- But employees fund then training themselves
  - Pay for it as long as the lifetime wage increase exceeds the cost of training